

REFLECTIONS ON THE GLOBAL PROJECT BASED LEARNING PROGRAM A JOURNEY OF INNOVATION IN TOKYO

By Bharathwaj Manoharan

Department of Electronics and Instrumentation

SRM Institute of Science and Technology

The Prelude Arriving in the Metropolis

The journey to Japan began long before the flight touched down at the airport. It started with a shared sense of anticipation and a deep understanding of the responsibility we carried as representatives of our respective universities. My colleagues and I decided to arrive in Tokyo a day early. This decision was driven by a highly practical need to understand the sprawling metropolis, learn how to navigate the intricate railway system, and sort out essential travel arrangements for our stay.

Stepping out into the crisp February air of Tokyo was a profound experience. The city operates as a masterpiece of modern engineering woven seamlessly with ancient traditions. That first day gave us a small but incredibly valuable window to settle in, recover from the fatigue of international travel, and adapt our internal clocks before the rigorous academic program officially began. We learned how to purchase transit cards, decipher the complex subway maps, and walk the orderly streets. This initial exploration was our first real test in collaborative problem solving, and it set a positive, proactive tone for the days to follow.

First Impressions Walking the Campus

Before the heavy lifting of the academic project commenced, we spent time exploring the Shibaura Institute of Technology Toyosu campus alongside the local university students who were participating in the program. This was a brilliant and organic way to break the ice. They kindly guided us through the massive library, the bustling student cafeterias, and the incredible modern architecture of the university buildings.

Walking alongside these students and sharing stories about our respective academic lives made the massive campus feel instantly welcoming. The atmosphere there was one of quiet dedication and a relentless pursuit of knowledge. Seeing the environment where they studied daily planted early seeds of inspiration in our minds. We realized that the standard for our upcoming project would be exceptionally high, and we felt a collective drive to rise to the occasion.

The Crucible of Ideation Forming Team Two

The program officially commenced on Thursday morning with a formal opening ceremony. The room was filled with sixty students from various corners of the globe including Italy, Thailand, Vietnam, Taiwan, Japan, and India. The air buzzed with a mix of nervousness and excitement. After a series of icebreaking activities that helped dissolve the initial awkwardness, the true challenge was unveiled.

We were grouped into multidisciplinary units, and I found myself assigned to Team Two. Our first task was to select a leader and confront our assigned project theme Reservation Energy and Conservation.

To be entirely honest, the topic felt overwhelmingly broad and somewhat limiting all at once. It seemed as though every conceivable idea in the realm of energy and conservation had already been heavily explored by established researchers and commercial industries. Finding a fresh, impactful, and technologically relevant direction felt like searching for a needle in a massive haystack. Furthermore, the program guidelines encouraged us to check all boxes within the theme, which added a significant layer of pressure to our initial brainstorming sessions.

The first couple of days in Room 406 were not easy. We stared at whiteboards, passionately proposing ideas and then meticulously dismantling them. We struggled to align our divergent thoughts and define a clear path forward. Our different academic backgrounds meant we often spoke entirely different technical languages.

However, what we did not realize in those moments of frustration was that this friction was the very foundation of meaningful collaboration. As we spent more hours together, debating and sketching out software architectures, we began to truly understand the strengths, perspectives, and unique working styles of each team member. Gradually, the noise settled into a productive rhythm. We learned how to align our diverse ideas and funnel our combined efforts toward a single, unified objective.

Deep Dive Real Time Embedded Control Systems

As our ideas solidified, we decided to tackle a critical issue at the intersection of climate change and food security. Our proposed solution was the Large Language Model Assisted Robotic Crop Health Monitoring System. To make this autonomous system viable, we had to rely heavily on the principles of real time embedded control systems. Because this is my specific area of academic focus, I want to elaborate deeply on how central this concept was to our entire project architecture.

A real time embedded control system is a dedicated computer system designed to perform specific tasks within a strict, predefined time constraint. Unlike general purpose computers that might delay a background task to prioritize opening a web browser or updating an application, a real time system must process inputs and generate outputs deterministically. If the system misses its deadline, the result is not just a delayed response, but a fundamental system failure. In engineering, we classify these into hard and soft real time systems. For agriculture, failing to water a crop during a critical heat spike acts much like a hard real time failure because the biological asset will sustain irreversible damage.

The relevance of this concept to modern agricultural automation is massive. Farming environments are highly dynamic. Traditional smart agriculture often relies on cloud computing, where field sensors send data to a remote server, the server processes the data, and then sends a command back to a water valve actuator. This introduces severe network latency. If the wireless connection drops during a severe heatwave, the crops could perish before the cloud reconnects to open the valves.

To eliminate this vulnerability, we engineered our project using localized embedded control. We partitioned our system into two distinct microcontrollers. The first was the Mobile Sensing Node built upon an Arduino Uno chassis. This rover was tasked with autonomous navigation through the crop fields. To ensure it did not crash into obstacles, it utilized a finite state machine and asynchronous hardware timers. Instead of using basic execution delays that would blind the ultrasonic sensors, the system polled the atmospheric environment asynchronously. This is a classic real time embedded programming technique ensuring that physical navigation always takes priority over data logging.

The second component was the Stationary Aggregation and Actuation Node utilizing an ESP32 microcontroller. This unit functioned as the central data sink and physical effector. This node embodied the absolute core of deterministic edge computing. We programmed a nine-state combinatorial matrix directly onto this edge device. The system evaluated the continuous analog data from soil moisture sensors against a hardcoded Boolean truth table. If the soil moisture dropped below a critical threshold while ambient temperatures spiked, the embedded system would instantaneously trigger a physical response. We simulated this rapid actuation with a buzzer alert during our prototype demonstration.

This edge layer guaranteed survival for the crops entirely independent of internet connectivity. The application of real time embedded concepts ensured our system possessed a robust failsafe mechanism. Understanding and implementing these rigid timing constraints is crucial not just for agriculture, but for aerospace

navigation, automotive antilock braking systems, and medical robotics. Bringing this level of deterministic control to our project was a major milestone for our team.

Integrating Artificial Intelligence

While the embedded hardware provided a reactive safety net, we wanted our entire system to be highly predictive. To achieve this, we developed a synchronous data acquisition pipeline. The ESP32 microcontroller concatenated the raw telemetry and the categorized matrix states into a rigid string of comma separated values. We bypassed traditional cloud middleware and streamed this data directly into a localized spreadsheet via a serial listening macro.

This approach was highly intentional and incredibly efficient. Large Language Models excel at processing structured tabular data. By logging directly to a spreadsheet, we automatically formatted the temporal data, creating a clean context window free of confusing formatting. We integrated the Gemini artificial intelligence model to serve as our secondary analytical layer.

The artificial intelligence ingested the structured data and performed advanced cross zone analysis. It correlated the spatial atmospheric data gathered by the rover with the subterranean moisture trends to generate nuanced predictive watering schedules and anomaly alerts. Seeing our software dashboard correctly identify severe stress indicators in specific zones based on our live sensor data was a moment of immense pride for the entire team. It was the culmination of our multidisciplinary efforts blending mechanical mobility, embedded control, electronic sensing, and cognitive cloud analysis.

Bridging Cultures and Building Bonds

The technical achievements of our week were substantial, but they represent only half of the story. The human element of this program was profoundly transformative.

Beyond the academic rigor and the endless lines of code, we connected on a deeply personal level. We learned about the respective cultures, interests, and personal backgrounds of everyone in Room 406. This exposure made the engineering experience significantly more enriching.

Language differences naturally posed a challenge. With team members from various linguistic backgrounds, English was sometimes not enough to convey a highly specific technical concept or a nuanced joke. However, technology helped bridge the gap. We frequently relied on translation applications like Papago. What could have easily been an insurmountable barrier instead became a wonderful

opportunity. We gathered around phone screens, reading translations, laughing at occasional misinterpretations, and finding common ground. It taught us to be patient, openminded, and deeply respectful of one another.

We made sure to explore Tokyo together outside of our laboratory hours. We navigated the vibrant, neon lit streets of Akihabara and absorbed the upscale elegance of Ginza. We experienced the pulsating energy of city life at night, sharing meals and swapping stories about our home countries.

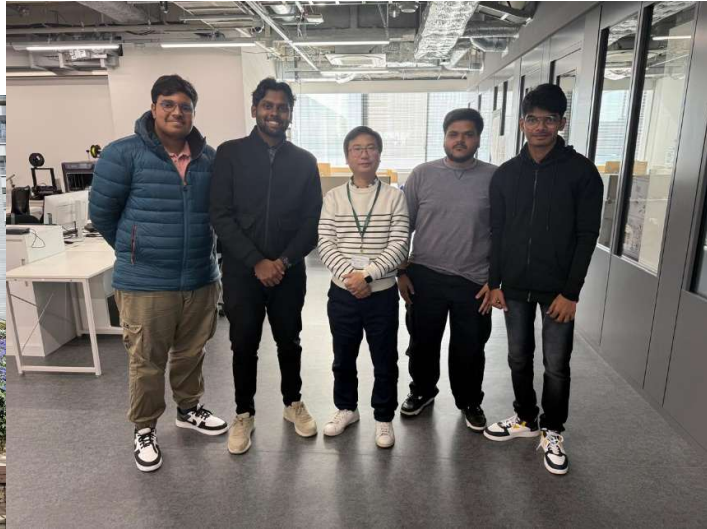
A particularly memorable excursion was our team visit to Miraikan, the National Museum of Emerging Science and Innovation. Standing together looking at exhibits dedicated to space exploration and the future of sustainable living, the broader purpose of our engineering education became crystal clear. We were not just learning to build machines we were learning to build a better future.

On Wednesday, amidst our frantic final preparations for the prototype deployment, we took a mandatory pause to participate in a traditional Yukata workshop. Stepping away from our laptops and donning the beautiful traditional garments offered a serene moment of mindfulness. It grounded us and provided a much-needed mental reset before the intense pressure of our final deliverables.

The Final Presentation and Laboratory Visits

Thursday arrived faster than any of us could have anticipated. From the morning until noon, we stood before our peers, professors, and mentors to present our final project. Demonstrating the robotic rover and explaining the complex data pipeline to the audience was exhilarating. We were not just presenting a technical prototype we were presenting the culmination of days of compromise, shared late night meals, and collective problem solving. The constructive feedback from the faculty, especially the valuable insights we had received earlier in the week from Professor Filippo Cavallo, validated our hard work.

With the intense pressure of the final presentation finally lifted, the program organizers arranged for us to formally visit a few key research laboratories at the campus as our week came to a close. We finally got to see the inner workings of the Biophotonics Lab and the Mobile Multimedia Communications Laboratory. Seeing these advanced setups at the very end of the program was incredibly meaningful. Having just built our own complex prototype from scratch, we possessed a newfound appreciation for the highly sophisticated research happening within those walls. Looking at their advanced optical setups and communication arrays tied our small-scale achievements to the broader landscape of global scientific innovation.



The Departure

The closing ceremony and farewell party that Thursday afternoon was bittersweet. As we reviewed our project summaries and received our certificates, I realized exactly how much I had grown. I arrived as an engineering student primarily focused on electronics, but I was leaving with a much broader understanding of global systems, artificial intelligence, and human collaboration.

I am incredibly grateful to have shared this journey with such an exceptional group of individuals. Our success was a direct testament to the diverse talents of our team

- Edoardo Tinti from Italy
- Sumirson Kumar from India currently pursuing his studies at SIT Japan
- Nahush Pethe from India
- Truong Van Thai from Vietnam
- Thamida Parhootit from Thailand
- Yuta Inoue from Japan

Before my scheduled departure on Friday morning, I took a solitary walk near the campus to reflect on the absolute whirlwind of the past week. Tucked away on a quiet street corner, bathed in the crisp morning sunlight, I stumbled upon a beautiful early blooming Sakura tree.

Standing beneath its delicate pink branches, I felt a profound sense of peace. The tree blooming vibrantly after a harsh winter perfectly mirrored our own journey starting with the cold friction of confusion and blossoming into a beautiful, functional collaboration. It was a quiet reminder of the beauty, the briefness, and the lasting impact of our time together in Japan. The technical knowledge I gained regarding embedded systems and artificial intelligence will undoubtedly shape my career, but the mutual respect, the cultural exchange, and the friendships forged beneath the Tokyo skyline are true treasures I will carry with me for a lifetime.